

CivicResolve: AI Public Grievance Management System

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Problem Statement



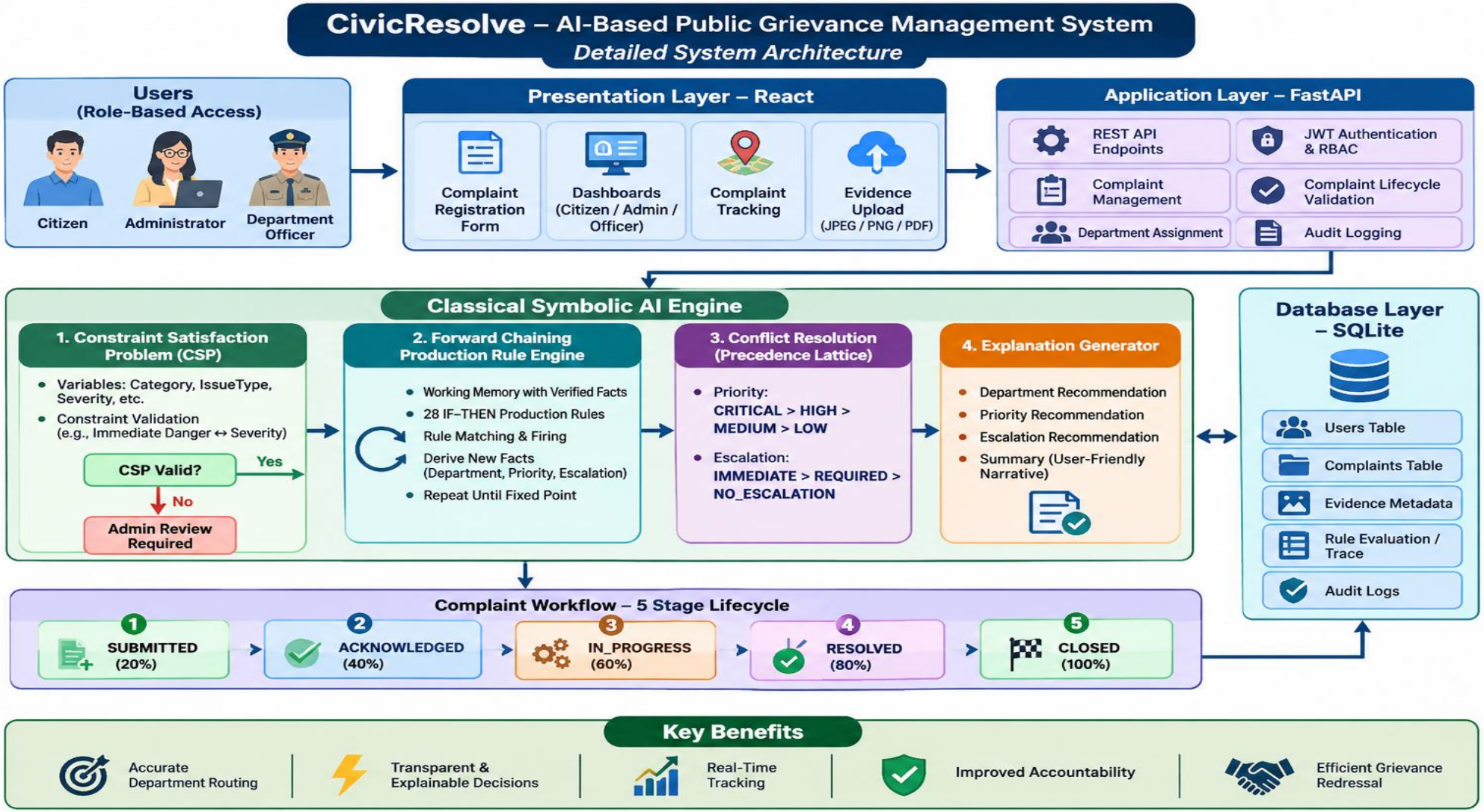
- Traditional public grievance systems often rely on manual complaint classification, department assignment, and prioritization.
- Complaints may be incorrectly routed or delayed due to inconsistent decision-making.
- Citizens face difficulties in tracking complaint progress and receiving timely updates.
- Limited transparency and accountability affect the efficiency of grievance redressal.
- There is a need for an intelligent, explainable, and systematic grievance management system.
- **Objective:** To develop an AI-based public grievance management system that validates complaints, recommends suitable departments, determines priority and escalation, and enables transparent complaint tracking.



Proposed System – CivicResolve

- A web-based public grievance management system with Citizen, Admin, and Department Officer roles.
- Uses Classical Symbolic AI for structured complaint validation and decision support.
- CSP validates complaint information and detects inconsistent inputs.
- Forward Chaining derives department, priority, and escalation recommendations using 28 production rules.
- Priority precedence resolves competing conclusions deterministically.
- Provides a five-stage complaint lifecycle with progress tracking.
- Maintains reasoning explanations, notifications, and an audit trail.
- **Key Advantage:** Explainable, rule-based AI with human administrative control.

Architecture Diagram



Algorithms Used



1. Constraint Satisfaction Problem (CSP)

- Represents complaint fields as variables, domains, and constraints.
- Validates category–issue mapping and detects contradictory inputs.
- Flags invalid complaints for Admin review.

2. Forward Chaining Production Rule Engine

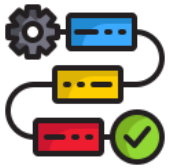
- Uses 28 predefined IF–THEN production rules.
- Initializes working memory with verified facts.
- Derives Department, Priority, and Escalation.
- Continues until no new facts are generated (Fixed Point).

3. Priority Precedence Lattice

- Resolves competing priority conclusions.
- Hierarchy: CRITICAL > HIGH > MEDIUM > LOW.
- Escalation: IMMEDIATE > REQUIRED > NO_ESCALATION.

4. Deterministic Explanation Generator

- Converts reasoning traces into readable explanations.
- Explains Department, Priority, Escalation, and Summary.



Complaint Workflow and Expected Outcome

Five-Stage Complaint Lifecycle

- SUBMITTED – 20% – Citizen registers complaint.
- ACKNOWLEDGED – 40% – Admin reviews and acknowledges.
- IN_PROGRESS – 60% – Department Officer starts work.
- RESOLVED – 80% – Officer completes the work.
- CLOSED – 100% – Admin verifies and closes.

Expected Outcomes

- Improved department routing and prioritization.
- Consistent and explainable AI decisions.
- Transparent complaint progress tracking.
- Department-level access control.
- Improved accountability through audit logs.
- Efficient coordination between citizens, Admin, and Officers.

Technology Stack: React | Python FastAPI | SQLite | JWT | bcrypt | Pytest